

**Identity Diffusion and Chinese Adolescent Mental Health:
A Sustainable Developmental Solution Based on the “Eggi” Digital Intervention**

Caddy Liu

New York University

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Dr. Danielle Key

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Background

Mental health problems among adolescents in China have drawn increasing scholarly attention. A meta-analysis found that the depressive symptoms affect 26.17% of Chinese adolescents (Zhou et al., 2024). A national survey reported that 79% of college students wished they had chosen a different major (103974, 2016), and Xu Kaiwen, former director of the psychological counseling center at Peking University in China, observed that around 40% of incoming students felt life was meaningless, which is a pattern he termed “hollow-heart syndrome” (2016). These figures point to a problem that goes beyond academic pressure. Many Chinese adolescents perform well academically but lack a clear sense of who they are and where they are headed. This causes them to not know why they exist.

Prior research has used Engel’s (1977) biopsychosocial model to examine how academic stress affects Chinese urban adolescents. Biologically, chronic stress disrupts hypothalamic-pituitary-adrenal axis functioning, raises cortisol levels, and impairs attention and memory (Stroud et al., 2009). Psychologically and socially, it is linked to contingent self-worth, perfectionism, and high family expectations. However, the biopsychosocial model leaves a key question unanswered: why do many adolescents feel empty and directionless even after academic pressure eases? This suggests that academic stress is not the root problem, it is a trigger for a deeper crisis.

That deeper crisis is identity diffusion. Erikson (1968) argued that adolescence is the critical stage for identity formation, during which young people work through questions like “Who am I?” and “What kind of person do I want to become?” If this process is blocked, the individual falls into identity diffusion. This marked by confusion, passivity, and a lack of stable self-

understanding. Further, Marcia (1966) distinguished four identity statuses: diffusion, foreclosure, moratorium, and achievement. In foreclosure specifically, some individuals commit to an identity without genuine exploration, such as accepting a career chosen by one's parents. The situation of many Chinese adolescents appears, on the surface, to resemble foreclosure. However, this paper argues that their compliance is externally imposed rather than internally integrated, showing a pattern better described as pseudo-foreclosure: outwardly conforming, inwardly empty; academically successful, personally absent. Once external pressures ease after college entrance exams, the suppressed need for identity exploration suddenly surfaces, but these young people lack both the skills and the support to respond. Identity diffusion follows. Xu Kaiwen's hollow-heart syndrome is a clinical expression of exactly this transition. **This paper argues that the mental health crisis among Chinese adolescents is, at its core, a crisis of blocked identity development. Effective intervention must go beyond reducing pressure and provide adolescents with a structured, developmental space in which to explore and form their identities.**

Structural, Cultural, and Relational Sources of Identity Diffusion

Identity diffusion is common among Chinese adolescents and does not imply weakness on behalf of the individual. On the contrary, this develops due to three overlapping pressures. Structural pressure: The Gaokao (College Entrance Exam) measures students on a standardized test that ultimately decides their lives post-high school including social class, career placement, and even marriage options (Zhou et al., 2023). The system only values students based on their academic performance. Zhou et al. (2023) found through qualitative analysis that this pressure consumes students' lives from morning until night with little room for identity formation.

Second, cultural conflict. Between traditional Confucian values that promote an other-

defined "relational self" rooted in filial piety and social harmony, and modern education's injunctions to students to "be yourself" and "follow your passion," adolescents are left unable to wholeheartedly embrace expectations placed upon them from without (Chen-Bouck & Patterson, 2016). Nor can they safely explore who they are from within. Third, lack of support. Mental health services in Chinese schools are underdeveloped, stigma about seeking help persists, and the vast majority of digital mental health interventions aim to treat symptoms rather than address needs related to development. The net result of these forces is that the psychosocial moratorium Erikson (1968) believed was necessary to healthy identity formation is being wiped out.

Limitations of Existing Interventions

While interventions exist across all three categories for adolescents' mental health specifically in China, there is a lack of interventions that adequately reach adolescents struggling with identity diffusion. Symptom-oriented interventions can help with anxiety and depression symptoms, but only target these manifestations of identity issues and not the underlying problems. For example, while an adolescent with identity diffusion may exhibit depression, treating that depression will not help them figure out "Who am I?" Interventions focusing on adolescents' sense of purpose have shown promise (Hielscher et al., 2023), but are still in their infancy and lack variation for cultural adaptation. Education-oriented interventions can reach students at a low cost, but simply teaching about mental health does not allow one to practice being mentally healthy. Therapy is the most professional option available, but there are extreme barriers to access, including a shortage of qualified professionals, urban-rural disparities, and stigma. Mental health interventions fall into these three categories, and all three lack the essential component of allowing adolescents to soundly and safely be unsure of who they are while supporting them in exploring who they could become. Developmental companionship fills that void.

Eggi: A Digital Companion System for Identity Incubation

This paper proposes Eggi, a digital companion application centered on identity incubation. The name is a variation of “egg” plus “i,” and its central metaphor is self-incubation: identity develops gradually in a suitable environment, much like a chick forming inside an egg. The shell cannot be forced open from outside. Similarly, identity incubation requires a safe, patient environment rather than external pressure or premature conclusions. Eggi is not a therapeutic tool nor a knowledge platform. It is a developmental space that allows uncertainty, accompanies exploration, and witnesses growth. Three principles guide its design: development over treatment (supporting a normal process rather than fixing a problem), process over answers (helping users observe who they are becoming rather than telling them who they are), and companionship over judgment (never criticizing users for feeling lost).

Eggi is structured to have three main modules. The former, Log Energy, challenges users to document any moments that lead them to feel alive and meaningful, not what they believe they should want. You can post a photo, voice message, or one keyword. It builds on findings of Pennebaker (1997) research on expressive writing which discovered that by consistently writing about emotional experiences one can significantly improve mental health. More to the point, Log Energy educates its users to pay attention to interior cues instead of exterior assessment criteria, slowly developing a sense of self-awareness. The second module, Method, assists the users to note role models in the areas that interest them with particular emphasis on patterns as opposed to outcomes. The user is allowed to select a domain that they are interested in and the system chooses people with varying backgrounds and stories of growth. This is not to copy what others have done but to see how others have managed to find what they love and how they went about it when it failed. This is based on the social learning theory introduced by Bandura (1977) and the concept of the zone of proximal development suggested by Vygotsky (1978): the person observes how others cope with the situation, which helps reduce the

anxiety of this person and provides him/her with strategies applicable in the future. The third module, *Emerge*, unites the records and observations into a visual personal narrative, a timeline or radar chart of energy moments and areas of interest over time. Users are able to rearrange this content, add reflections, and create their own life story. This design is based on the narrative identity theory proposed by McAdams and McLean (2013), which states that identity is constructed through the process of continuously telling and revising the story. *Emerge* is a low-barrier tool to this process, which helps users slowly come to feel a sense of authorship over their own lives.

Aligning with UNESCO's Education for Sustainable Development (ESD) Framework

The phenomenon of developmental crisis of identity diffusion among Chinese adolescents is not an isolated occurrence but a local expression of a global issue: how do individuals develop a coherent, agentic sense of self in the face of rapid social, technological, and environmental change? This is the core issue of the Education for Sustainable Development (ESD) towards 2030 framework by UNESCO. ESD is an educational framework that is comprehensive and aims to empower people to reflect on their actions considering their current and future social, cultural, economic and environmental consequences, in both a local and global perspective (UNESCO, 2020). The primary objective is to assist people in becoming a self-directed team of individuals who will have the capacity to establish their own values and sense of agency to become responsible individuals who can handle complex situations and make the world a better place.

Eggi's design is directly aligned with the core competencies and principles of ESD.

To begin with, ESD focuses on the creation of major competencies such as critical thinking, systems thinking, self-awareness, and collective action (UNESCO, 201). These competencies in the identity development of adolescents are operationalized by Eggi in three modules. *Log Energy* develops self-awareness through training users to monitor the inherent motivations in themselves. *Method*, based on understanding role models, promotes critical thinking and anticipatory thinking by

encouraging users to analyze various life paths, and not accept a single life path, which is imposed on them. Emerge supports the synthesis of experiences into the coherent life narrative, answering the question Who am I in relation to the world I want to live in? supports both the synthesis of experience into the coherent life narrative, and normative competence (reflecting on what is important to one) as well.

Second, ESD facilitates lifelong learning and life-wide learning that goes beyond formal schooling into non-formal and informal learning environments (UNESCO, 2020). In the case of adolescents caught in the Gaokao system, the formal curriculum does not leave room to explore one's identity. As a digital companion, Eggi offers exactly this missing legitimate moratorium, a safe and informal learning environment, where developmental exploration can be achieved at the own pace of the user, without being constrained by the school schedule or the examination. It is an embodiment of the ESD ideal that transformation learning has to be a continuous self-directed process.

Third, the approach of creating learning cultures adopted by the whole institution, promoted by ESD, is encouraged to develop the values of sustainability. The so-called phenomenon of hollow-heart syndrome is one indicator of a failure of the educational institution to help nurture the inner life of its students. Eggi does not react by criticizing the system per se, but instead provides a complementary, learner-focused digital ecosystem in which the values of autonomy, reflection, and purpose are put into practice on a daily basis. In this sense, Eggi can be viewed as productized innovation that will deliver the core goals of the ESD for 2030 framework-building human agency and transformative capacity- in response to the specific crisis of meaning and identity in the digital age.

Developmental Rationale and Ethical Boundaries

The design of Eggi is based on four theoretical mechanisms. To begin with, it provides a

legitimate moratorium (Erikson, 1968): a cheap, virtual space where users can play around without having to defend themselves to anyone. Second, it serves as a safe haven (Bowlby, 1969): everywhere, without judgement, without disappointment or derision, providing users with the security they need to go out there. Third, it facilitates construction of narrative identity (McAdams and McLean, 2013) by using daily micro-records, which naturally builds up to a coherent story of the self, without requiring its users to write out a full autobiography at once. Fourth, its game-based, low-entry interaction design (Brown et al., 2016) one energy moment per day, visible timeline growth, data complete control by the user, etc. solves the adherence issue that plagues most mental health interventions.

There are a number of ethical limits that are crucial. Eggi is not a medical device. In the case of adolescents who already have severe depression, self-harm or suicidal thoughts, Eggi has an in-built crisis detection system that presents local hotlines and makes users seek professional help. The user owns all their data: locally by default, and Eggi does not use personal data to conduct commercial or research activities without explicit consent. Lastly, Eggi is a design having a graduation mechanism. The app promotes independence and not dependence after a few months of use, the system actively encourages users to apply their growing skills in real life by sharing energy moments with their friends, identifying role models in their real communities, so that the app is supporting independence and not dependency.

Implementation and Evaluation

Eggi can be implemented in three environments. It may be used in conjunction with career planning and mental health courses in schools as a tool that the student can use to continue to employ himself or herself in self exploration beyond the classroom. It inspires a meeting middle way between parent and child in families: instead of addressing each other with the question

What do you like? the children can choose the timeline or energy moments she or he wishes to share with parents, thus making the conversation more natural and authentic. Eggi may also serve as a primary developmental aid prior to the escalation of problems in a situation of Eggi functioning in a rural setting that lacks adequate resources to sustain the provision of professional adolescent mental health services, which is nearly non-existent in under-resourced rural areas.

A mixed-methods approach is suggested to measure the effectiveness. The quantitative aspect would consist of a pre-post control group design; thematic variables would include identity status (EOM-EIS-II) and depressive behaviors (PHQ-9), sense of meaning (MLQ), intrinsic motivation in an Eggi group (three months of use) and a waitlist control group. The hypothesis is that the Eggi group would significantly higher score on identity exploration and a sense of meaning and significantly lower the score on identity diffusion as compared to controls. The qualitative element would entail semi-structured interviews and examination of freely provided Log Energy records as the years come by in an effort to trace how narratives of identity were formed over the years. A transition to any recording of the external event (I got a high score today) to internal experiences (I tried to paint the first time) would be indicative of identity incubation in progress. Whether the effects of Eggi may vary among the populations of cities and town as well as among different students with different academic backgrounds, should also be addressed by future research in order to refine its design and provide equitable scaling.

Conclusion

In this paper have begun by posing the following question: why then do the majority of the academically successful Chinese adolescents walk out of the university building empty and lost? The solution is that they are conditioned themselves to become best test-takers of their own lives

but not be given the opportunity to become writers of their own lives. This identity-diffusion crisis, as discussed in this paper, has a local expression of global challenge that Education for Sustainable Development (ESD) framework by UNESCO attempts to address: how do we go about developing agentic and responsible individuals to navigate the complexity and shape a sustainable future?

It is in that spirit that Eggi, the digital companion system, the identity incubation of which will be the foundation of this paper is the subject matter of this paper. Eggi offers an opportunity to adolescents to find themselves in a safe, non-judgmental and sustainable space with three modules: Log Energy, Method, and Emerge. The design proposed by Eggi, which is critical, incidentally, is a direct operationization of the ESD 2030 framework: "Log Energy" develops the self-awareness, "Method" train the critical and anticipatory thinking, and finally, is the emerging process that supports the development of the narrative identity and normative competence; thus, the suggested design is critical. It reflects how digital advances can make the lofty objectives of worldwide educational methods into their practical, reachable application. Another voice is not needed by the Chinese teens and even around the world, which might enlighten them on what to do. They need a place in which they can be unsure, which will come with them as they explore, and they will see development. Eggi is conceived as such space: a digital friend to becoming a sustainable self in an ever-changing world.

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